



GROUP PROJECT (4)

OUR TEAM

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RIDE-HAILING ANALYSIS

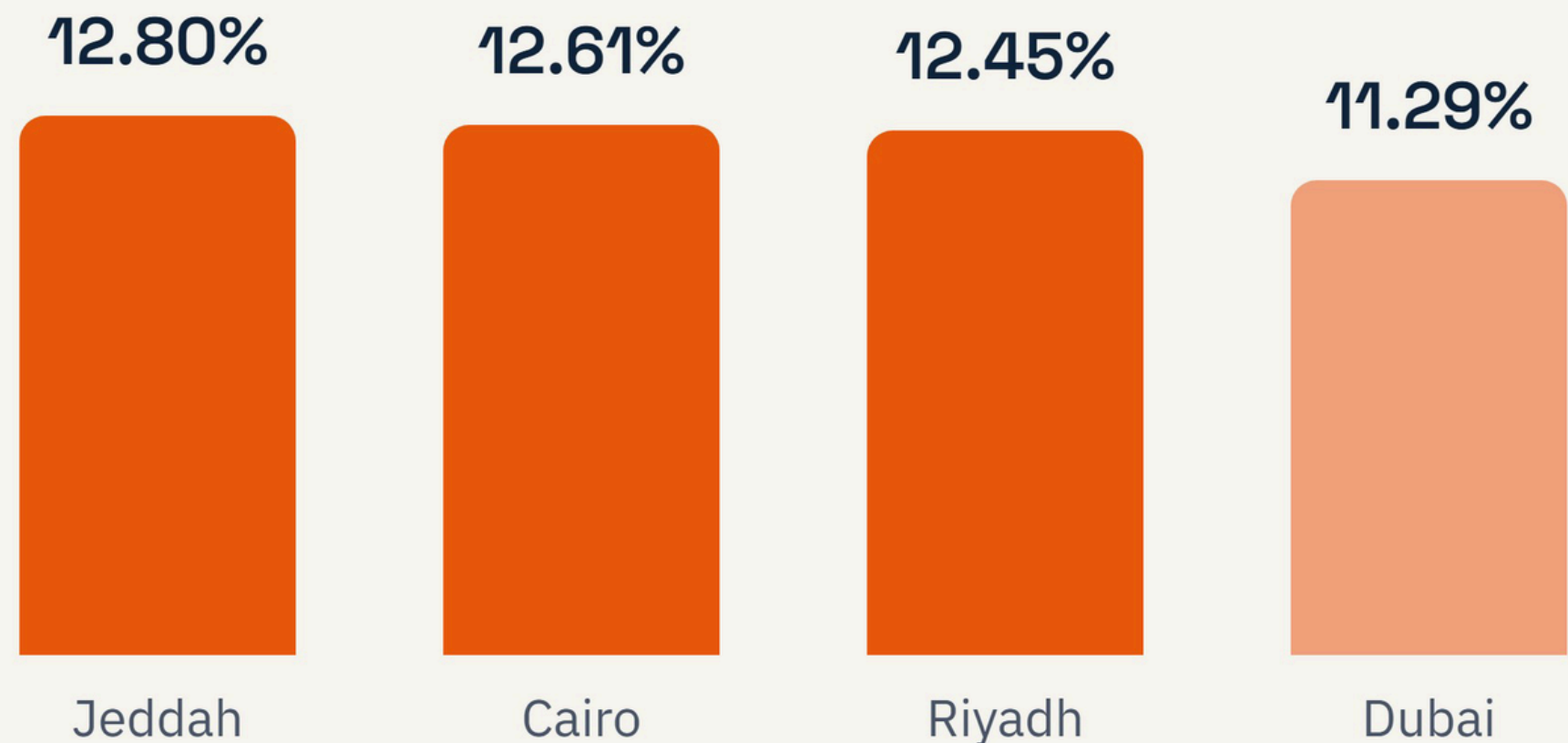
Surge Pricing, Trip Demand, Weather & Driver
Behavior
SQL Analysis Project

BUSINESS CONTEXT & OBJECTIVE

- The platform tracks rides, drivers, weather, and driver behavior.
- Weather and traffic can affect trip duration, pickup ETA, surge pricing, and cancellations.
- Objective: use SQL analysis to answer 15 business questions and identify opportunities to optimize pricing and driver supply.

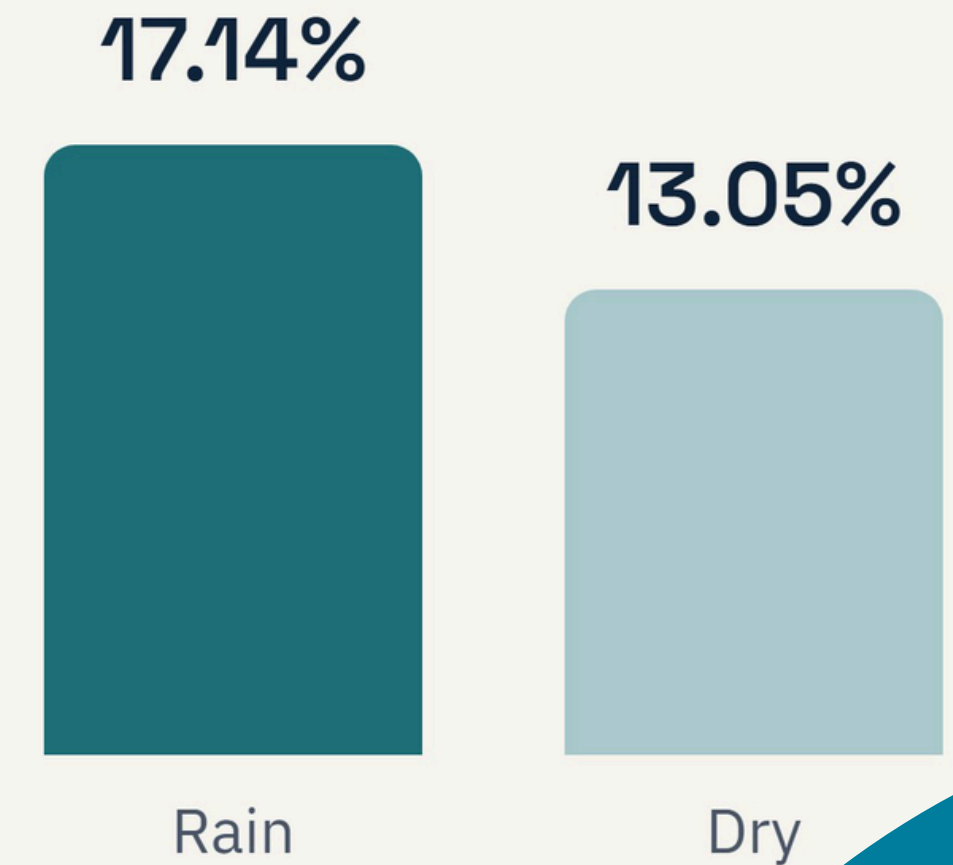
Key Insight 1: Cancellations

- Jeddah had the highest cancellation rate: 12.80%.
- Cairo: 12.61% | Riyadh: 12.45% | Dubai: 11.29%.
- Rain increased cancellations from 13.05% to 17.14% — about 4 percentage points.
- Most common reasons: Rider cancel (613), No driver (600), Driver cancel (502).
- High winds did not show higher cancellations in this dataset: 2.56% vs 13.74% during low winds.



Key Insight 2: Weather Creates Operational Pressure

- Rainy trips averaged 15.25 minutes vs 14.83 minutes when dry.
- Rain increased pickup ETA from 4.0 to 5.5 minutes.
- That is a 1.5-minute increase, or about 38%.
- Main takeaway: rain has a stronger visible effect on ETA and cancellations than wind in this dataset.



Key Insight 3: Surge Pricing Patterns

- 61.77% of all rides had surge pricing (12,353 out of 20,000 rides).
- Evening had the highest average surge multiplier: 1.09.
- 7 PM was the peak hour, with an average surge of 1.21x.
- Surge rides had longer pickup ETAs: 270.37 sec vs 238.67 sec.
- Average fares generally increased as surge multipliers increased across cities.

SURGE RIDE SHARE

61.77% (12,353 out of 20,000 rides)

PEAK SURGE MULTIPLIER

1.21x at 7:00 PM (Evening Avg: 1.09x)

Pickup ETA: Surge vs. Non-Surge

Non-Surge Rides

238.67s

Surge Rides (+13.3% Longer)

270.37s

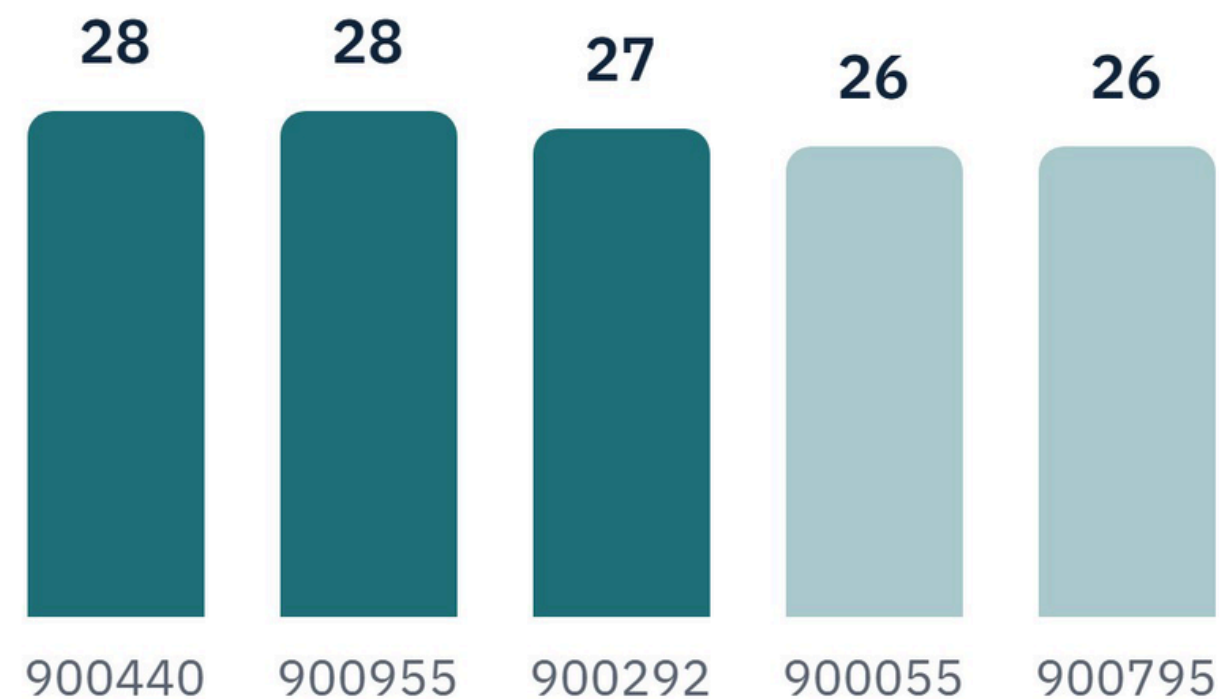
Key Insight 4: When Demand Peaks

- 6 AM had the highest trip volume: 896 trips.
- Evening demand was also very strong:
- 5 PM: 864 | 6 PM: 865 | 8 PM: 881 | 9 PM: 881
- Demand is concentrated around early-morning and evening commute periods.
- Operational implication: driver supply should be closely managed around these peak windows.

Key Insight 5: Driver Performance

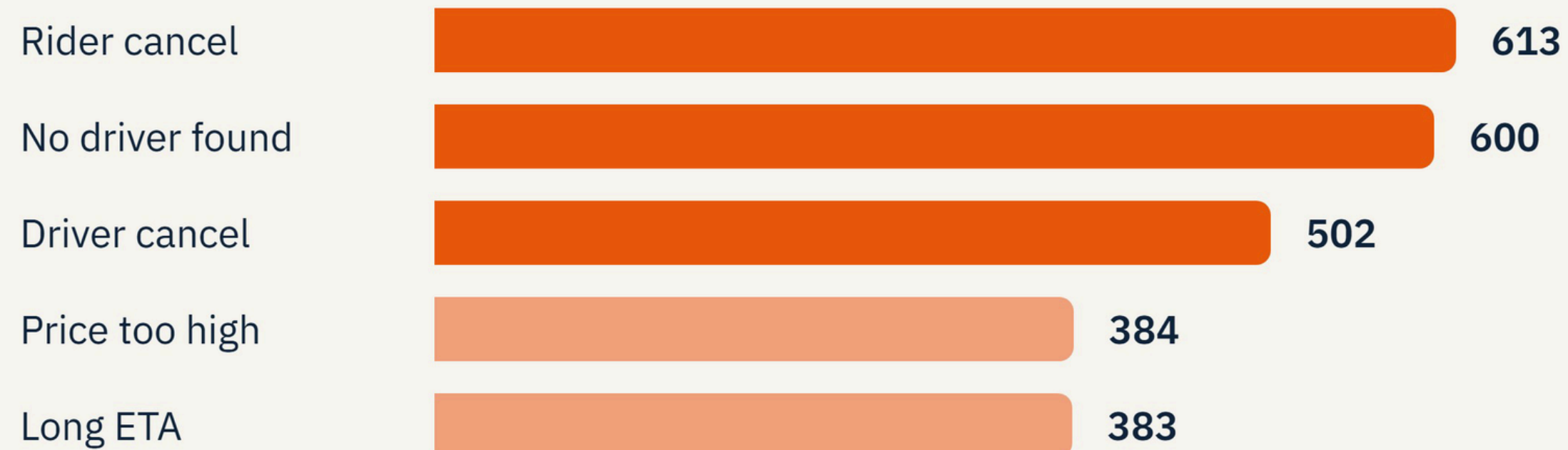
- Top ride-volume drivers: IDs 900440 and 900955 with 28 completed rides each.
- Top revenue driver: ID 900463 with \$30.27.
- New drivers averaged 14.65 completed rides; older drivers averaged 14.54.
- Difference: only 0.11 rides — driver tenure alone was not a strong predictor of ride volume.

Top drivers by completed rides



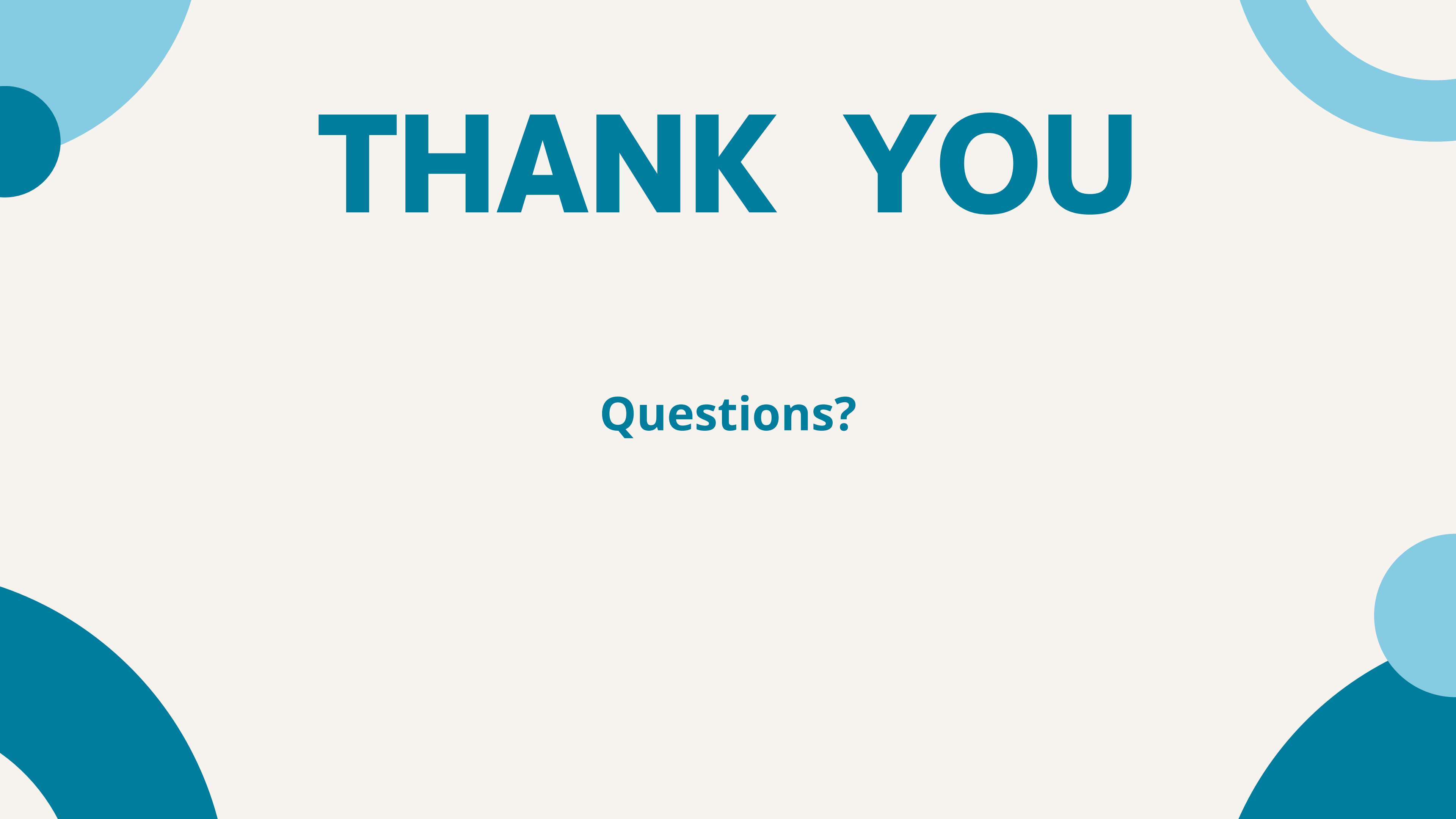
Additional Finding: Fares by Ride Outcome

- Cancelled rides averaged \$4.53.
- Completed rides averaged \$4.50.
- The difference was negligible.
- This suggests fare level alone did not meaningfully distinguish cancelled and completed rides in this dataset.



Management Recommendations

- Prepare additional driver supply during 6 AM and 5–9 PM demand peaks.
- Use weather-aware operations during rain to reduce pickup delays and cancellations
- Monitor 'No driver' and rider cancellations, the two largest cancellation categories.
- Review surge strategy during evening peaks, especially around 7 PM.
- Focus driver interventions on availability and operations rather than tenure alone.

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THANK YOU

Questions?